Lesson notes · for you

Loss, descent and backprop

University, 2

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- Loss is large when wrong
- Step against the slope
- Backprop passes the blame back

Activity: In the lab

1. Run Good
2. Run Too big

Discussion

- Where does the gradient point?

Check yourself

- θ^2 at 3, rate 0.1: write the next θ.

Try it this week
Compute the 2.4 step on paper without looking it up.

Worksheet

Loss, descent and backprop

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Name _____ Class _____ Date _____

1 **Loss θ^2, $\theta = -2$, $\eta = 0.25$. θ after one step?**

◯ -1 ◯ -3 ◯ -1.5

2 **The gradient points at the minimum.**

◯ True ◯ False

3 **Match the curve to the cause.**

Loss grows every step	• • • •	Step too small
Loss falls very slowly	• • • •	Memorising the training set
Train loss low, new data worse	• • • •	Step too big

4 **Order one descent step.**

☐ Compute the gradient

☐ Step against the gradient

☐ Compute the loss

5 **$y = w \times x$, loss $(y - t)^2$, $x = 2$, $w = 1$, $t = 4$. What is the loss's slope with respect to w?**

◯ -4 ◯ 8 ◯ -8

✎ **What does a growing loss tell you, before you look at the weights?**

Check your answers

Loss, descent and backprop

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1 **−1**

The slope is $2\theta = -4$. Step against it: $-2 - 0.25 \times (-4) = -1$, closer to the bottom at 0.

2 **False**

It points uphill, locally. Descent steps against it.

3 **Loss grows every step — Step too big**
Loss falls very slowly — Step too small
Train loss low, new data worse — Memorising the training set

The learning rate and overfitting leave different fingerprints on the curve.

4 **Top to bottom: 2, 3, 1**

You need the slope before you can step against it.

5 **−8**

Chain rule: the loss's slope with respect to y is $2(y - t) = 2 \times (2 - 4) = -4$, and y's slope with respect to w is $x = 2$, so $-4 \times 2 = -8$.